

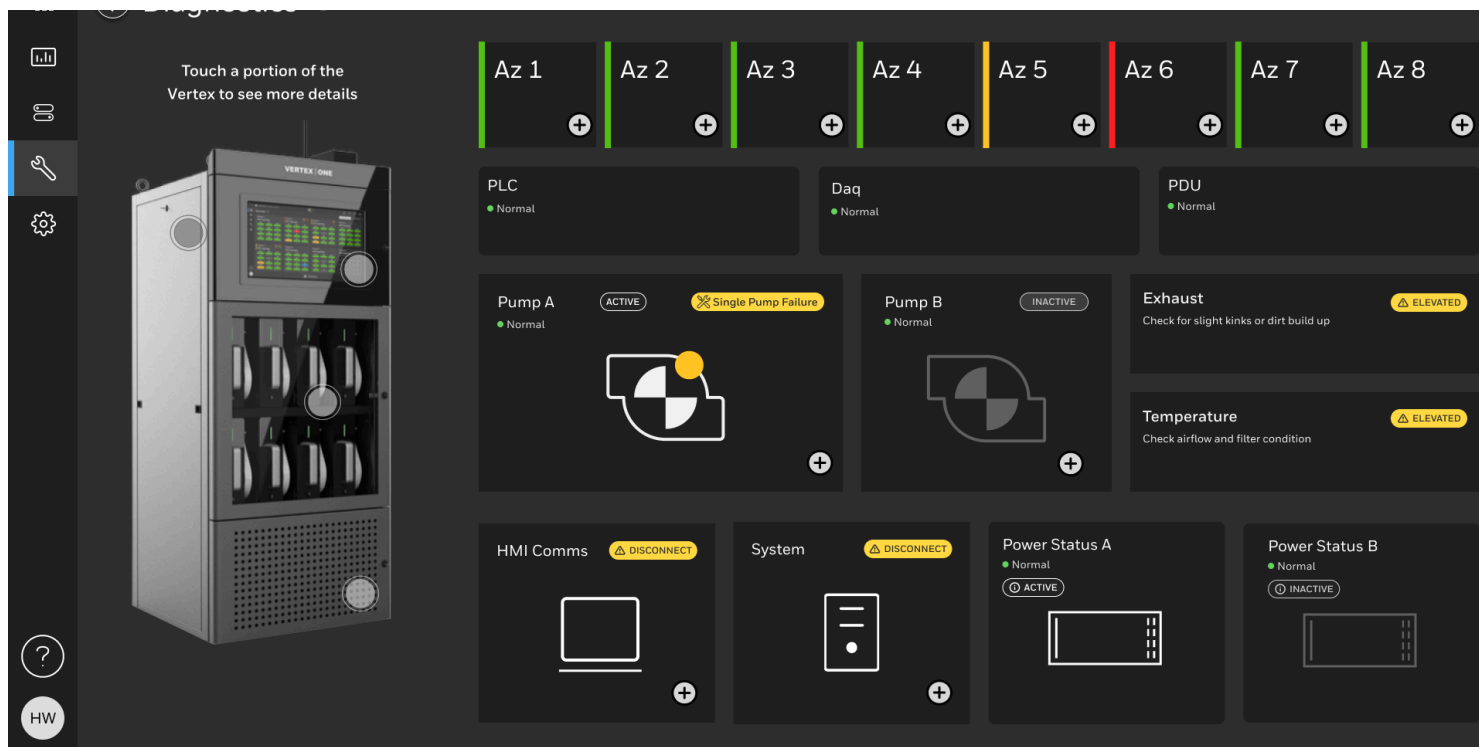
Matt McGlothlin

VertexONE - Gas Detection in Semiconductor FABs

As the UX Lead for the Vertex One program, I led the end-to-end design of Honeywell's next-generation toxic gas monitoring system for semiconductor fabs—an evolution of the award-winning Vertex Edge platform. The project integrated hardware, embedded interfaces, and cloud-ready HMI software to improve serviceability, ergonomics, and usability in high-containment environments.



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End-to-end Design

As UX lead for the project I was responsible for delivering a modern experience aligned with Honeywell's latest design system. This included working across the product team spectrum with developers, engineers, product and project management, sales and marketing. My teammate on the award-winning Honeywell User Experience (HUE) included a user researcher and project manager. Together we tackled the problem by following a proven design process starting with discovery – desktop research, stakeholder interviews, competitive analysis, observational, virtual voice of customer (VOC) and workshops that help uncover root causes to redundant problems. Themes start to emerge as I synthesize our learnings and distill them down into common themes. Artificial intelligence tools aid efforts to pinpoint patterns in the findings. I'm able to prioritize potential features while transitioning into the next phase delivering mapping current screens, creating optimized workflows, wireframes and prototypes to validate and test with customers and stakeholders. Artificial intelligence is increasingly sprinkled into this design phase to aid creativity and efficiencies. Like a photo slowly coming into focus, designs and concepts come into view.

Throughout this stage I'm thinking about working within the latest design system as well as any business constraints. – bearing in mind technical feasibility and deadlines while balancing the innovation required to build a differentiating experience streamlining, improving existing workflows. Often times this required choreographing the experience with engineers and industrial designers to ensure the software and hardware work in harmony with digital controls and guided steps guiding the user through complex setup, maintenance and troubleshooting procedures. The result is machine and man working towards the a common goal of preventing expensive downtime in the semiconductor FAB. I worked with design system peers to contribute new components that work for a large tablet-like touchscreen. The conceptual prototype experience received enthusiastic support from field service technicians, stakeholders and customers.

With concepts finalized and prototypes testing, I prep for dev hand-off finalizing and documenting designs. Iterative design and complexity can lead to a chaotic development cycle and frustrated designers.

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smoothing.

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VERTEX ONE

Key HMI Differentiators

ALARM/FAULT INTELLIGENCE

Analytics and insights with modern data visualization



Future Release

DIGITIZE MAINTENANCE

Embedded AI assistant to reduce ticket escalations



Future Release

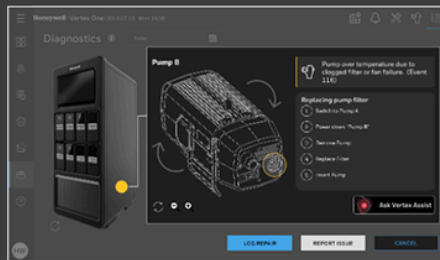
DOWNTIME PREVENTION

Predictive maintenance with AI-generated insights



SIMPLIFY LABOR INTENSIVE TASKS

Advanced diagnostics with guided steps and realistic diagram



FLEXIBILITY (MVO 2)

Remote management of multiple Vertex from various devices.



Honeywell

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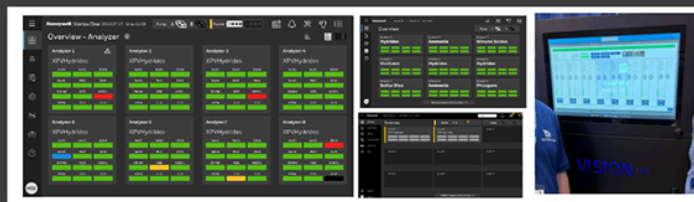
26

VERTEX ONE

Key HMI Enhancements

OVERALL EXPERIENCE

Continuous UI improvement (Edge > C > One) strengthens Honeywell usability advantage over major competitors like DoD.



CYBERSECURITY

Auto-logout activates idle screen highlighting Vertex status.



USABILITY

Point mapping system increases productivity and improves recognition/recall of point location configurations.



INTEGRATION

TGMS inhibit timer enables FSE to prevent automatic faults



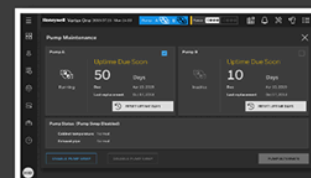
DATA INTEGRITY

Backup and restore prevents data loss and reduces downtime due to unforeseen conflicts.



SAFETY

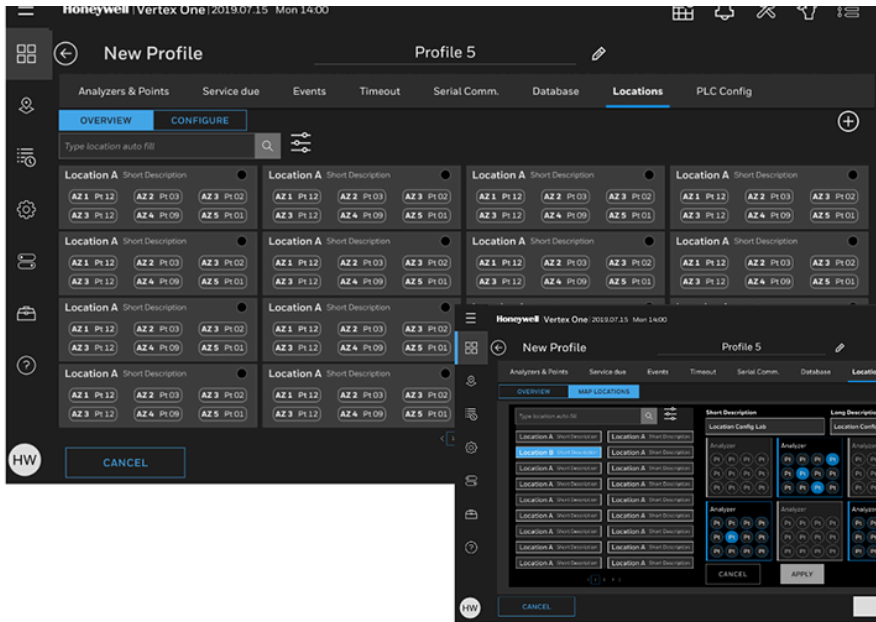
Disabling auto pump swap prevents potential safety issue when working on backup pump.



Honeywell

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FEATURES

Card-oriented Layout

Reduces table layout overload and makes the UI more touch friendly

Point Mapping

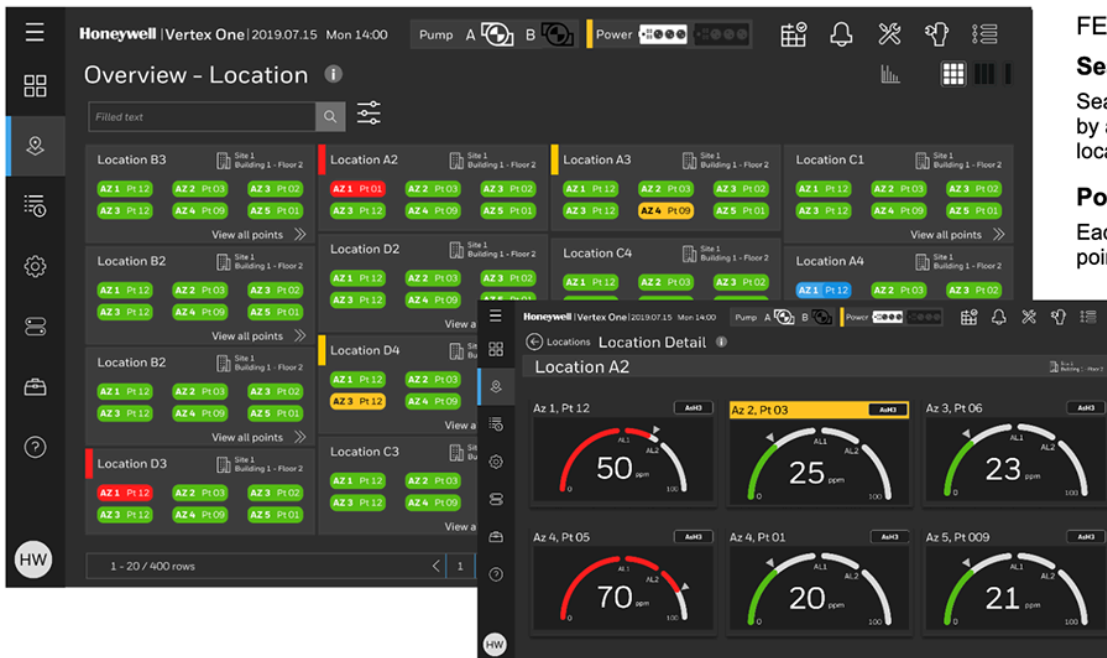
Consistency of point mapping since the interaction pattern is similar to PLC config. Therefore user only has to learn the functionality one time.

Search and Filter

Search and filter locations by short description, long description, AZ and PT

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LOCATION GROUPING



FEATURES

Search, Sort and Filter

Search for a specific location or filter by alarm, fault, building. Sort by location name or location status

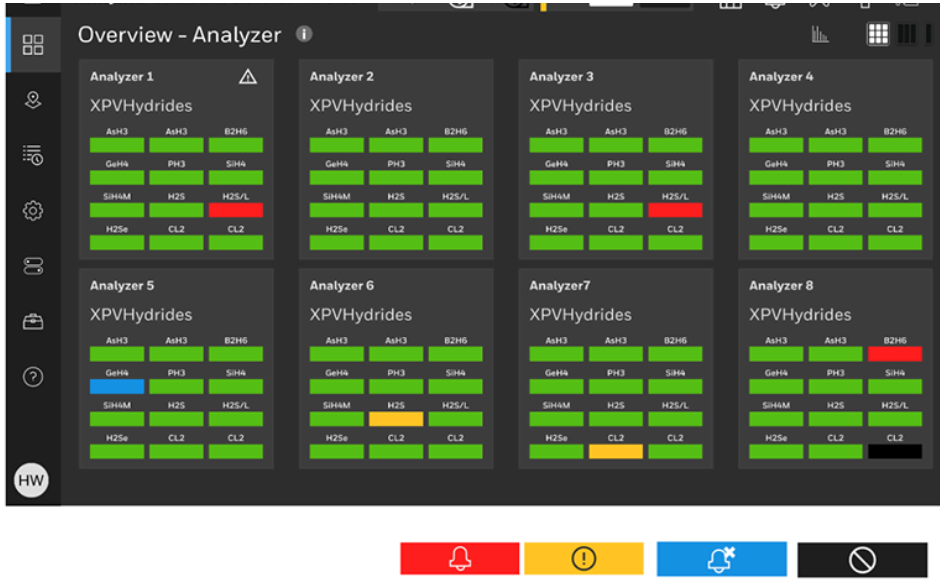
Point mapping

Each locations displays up to six points with Analyzer # and Point #

Gauge Charts

Location detail displays gauge chart with alarm thresholds, concentration and any faults highlighted in yellow.

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Power Status

Header iconography allows user to monitor pump and power status in one glance. Selecting either graphic enables user to manage both.

Legend

New users unfamiliar with the UI can touch info icon for legend that defines point color and icons.

Layout Switcher

Switching between grid, column and single analyzer view gives users more flexibility and options for viewing analyzer status

Icon-assisted Points

Color blind users are able to distinguish between red and green easily with the help of icons

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REMOTE MANAGEMENT (MVO2)

FEATURES

Multiple Vertex Overview

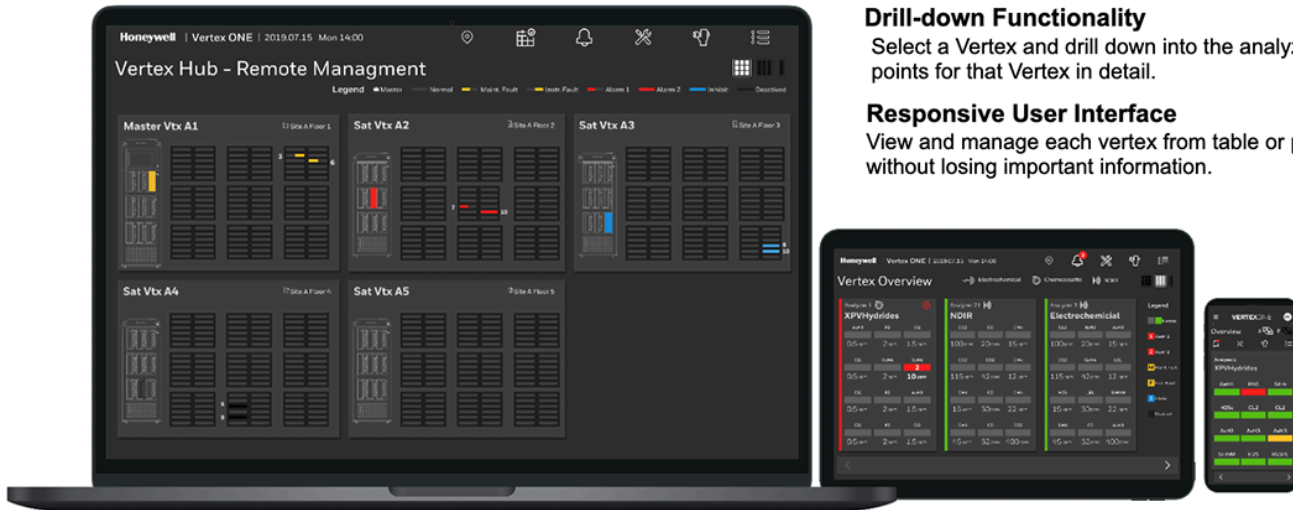
See status of all analyzers/points on all Vertex in one screen including alarms, faults and inhibits.

Drill-down Functionality

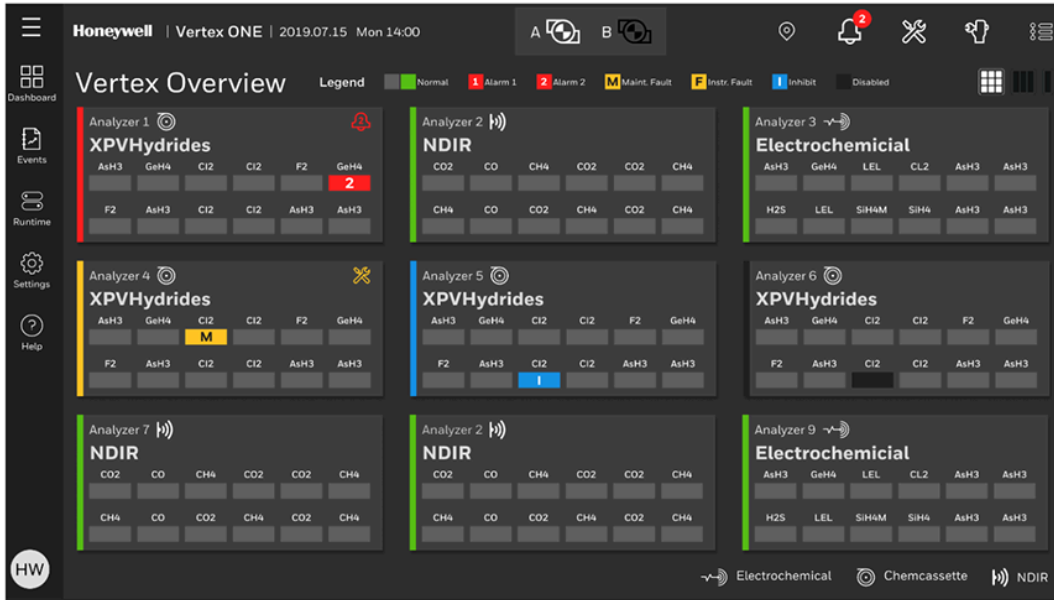
Select a Vertex and drill down into the analyzers/points for that Vertex in detail.

Responsive User Interface

View and manage each vertex from table or phone without losing important information.



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Iconography

NDIR, Chemcassette and ECC icons signify to user which sensor analyzer contains

Legend

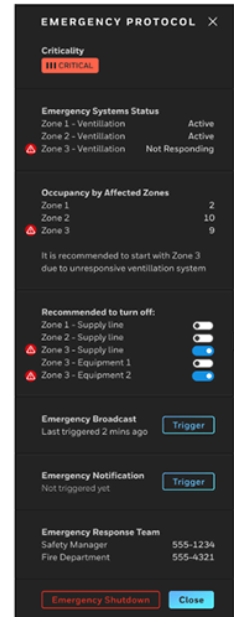
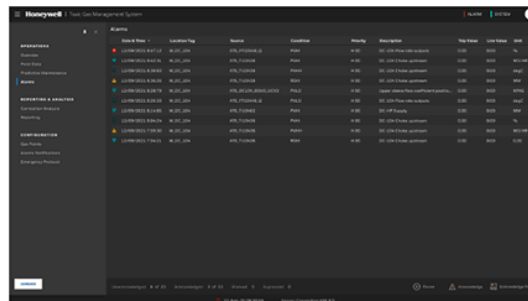
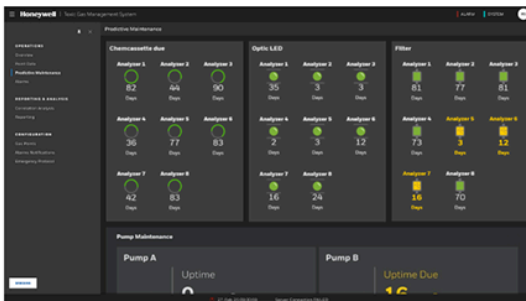
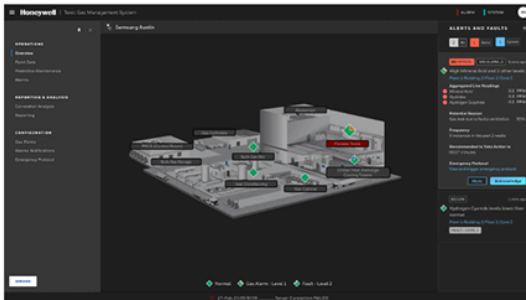
Bottom legend displays definitions for each type of sensor when selecting.

Sensor management

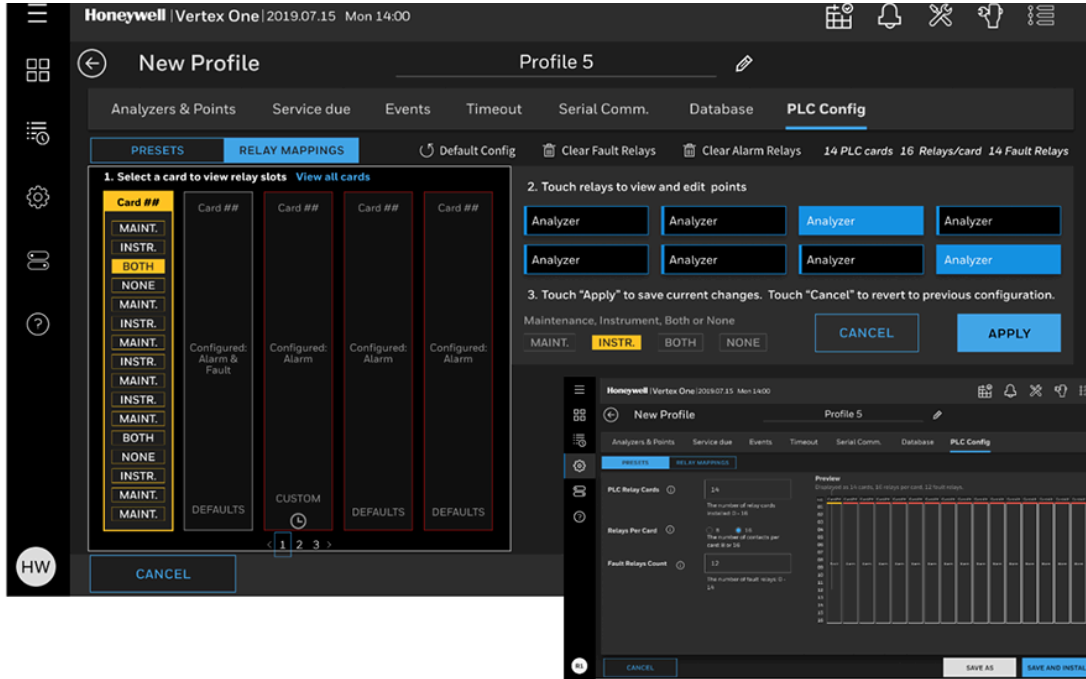
Bump and calibration process for ECC and NDIR sensors similar to Midas M.

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TGMS INTEGRATION (WIP)



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FEATURES

Presets and preview

User can visualize the card preset settings including type and amount as well as interact with tooltips that explain terminology = More efficiency.

Guided Relay Mapping

Each step is activated in a just-in-time fashion so that the user is never overwhelmed with choices.

Forgivable interface

Every step is reversible in case of mistake and changes are denoted with pending icons so user never loses track of changes.

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IDLE SCREEN (WIP)



FEATURES

Auto-logout

Customer is able to configure the logout timeout as well as enabling feature.

At-a-glance status

Customer can view status from a distance and quickly gather information on the current status of Vertex.

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OVERALL EXPERIENCE

Continuous UI improvement (Edge > C > One) strengthens Honeywell usability advantage over major competitors like DoD.

USABILITY

Point mapping system increases productivity and improves recognition/recall of point location configurations.

DATA INTEGRITY

Backup and restore prevents data loss and reduces downtime due to unforeseen conflicts.

INTEGRATION

TGMS inhibit timer enables FSE to prevent automatic faults

SAFETY

Disabling auto pump swap prevents potential safety issue when working on backup pump.

Touch to login

Touch to login

Touch to login

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Vertex One Design Workshop – Summary and Outcomes

Held at Honeywell’s Lincolnshire headquarters, the Vertex One UX Workshop brought together cross-functional stakeholders from product management, engineering, human factors, and field services to align on the next-generation Vertex One vision. The session focused on reimagining the human-machine interface (HMI) and overall user experience for toxic gas detection in high-risk semiconductor FAB environments. As the UX Design Lead, I helped shape the workshop agenda, methods, and synthesis framework that guided subsequent concept validation at Samsung Austin. The workshop surfaced key challenges in integrating and maintaining Vertex systems, with emphasis on cybersecurity, preventative maintenance, documentation, system upgrades, and service efficiency. Discussions included field service team members for Intel, Samsung, and other partner accounts.

Prework and agenda

The technical support performance analysis revealed key factors affecting customer experience and service efficiency for the Vertex product line. The study identified documentation, software support, and analyzer troubleshooting as the top drivers of customer assistance requests.

Key insights:

- **Documentation issues** account for the highest number of tickets (121) and the most accessed knowledge articles (100+).

Attendees



Prework - Customer Support Tickets

Category	Count
Documentation	121
Software Support	85
Analyzer Troubleshooting	72
Hardware Issues	45
Integration	38
Security	25
Performance	18
Compliance	12
Other	10

Category	Count
Documentation	121
Software Support	85
Analyzer Troubleshooting	72
Hardware Issues	45
Integration	38
Security	25
Performance	18
Compliance	12
Other	10

Agenda - Day 1

Day 1 Session 1 (Jeremy, Steven)

- End to end services
- Samsung Taylor visit with Richard notes

Day 1 Session 2 (Erik, Tao, Siegal)

Design details for Vertex One AZ and Rack systems

- Where we are at with architecture
- Where we are at with global PDU
- Modularity of major elements

Feedback - What is meant by modularity?

Optics module could be modular	Modularity means it's easy to swap out	Modularity means it's easy to swap out
Modularity means it's easy to swap out	Modularity means it's easy to swap out	Modularity means it's easy to swap out

CTO Guidance

CTO Guidance: The CTO should ensure that the Vertex One system is designed to be modular, allowing for easy integration and maintenance. This will help reduce downtime and improve overall system reliability.

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- Hardware-related searches indicate a strong reliance on replacement parts knowledge, particularly for optics, pumps, and PLC components.
- Top knowledge-base searches—**software installation, optics maintenance, flow and fault codes**, and calibration—mirror the main escalation categories.

Together, these patterns show that user friction peaks during installation, configuration, and maintenance workflows—especially around software compatibility, optics calibration, and documentation clarity. Streamlining these areas with **improved procedural guidance, interactive digital tools**, and **enhanced remote diagnostics** could significantly reduce escalation rates and technician downtime.

Workshop sessions

Over four collaborative workshop sessions, a cross-functional team—including field service engineers, tech support leads, software developers, mechanical and electrical engineers, systems engineers, offering managers, and global fellows—examined the end-to-end customer experience of Honeywell’s gas detection systems.

Day 1:

Discussions focused on service inefficiencies, training gaps, and nuisance alarms in semiconductor fabs. Human factors were central, referencing Fitts’ Law, ergonomics, and Nielsen Norman heuristics to assess HMI usability. Microsoft Copilot supported synthesis by summarizing transcripts and quotes. A review of the Samsung customer visit highlighted repair turnaround challenges, pump reliability, and system architecture issues, leading to analysis of **PLC configuration**, alarm logic, modular design, and HMI responsiveness.

DAY 1 - Session 1



Quotes from transcript (Copilot Assisted)

Quotes from transcript (Copilot Assisted)



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improve **training and diagnostics**, reduce false alarms, and **predict failures**. Key opportunities included improving **Chemcassette maintenance**, parts replacement, software releases, PLC and pump duty cycles, and **predictive tools** to drive reliability and service excellence.



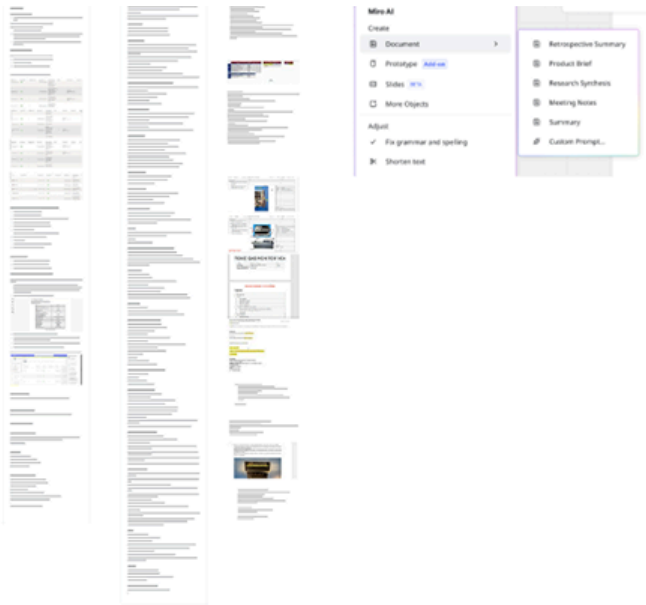
Quarterly report for customers and notes



Meeting Notes - Assisted with MS Copilot



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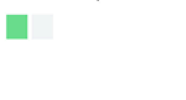
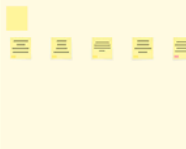


Wireframes



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Workshop 1: Day 1

Automate Reports? ★ 	How do we decrease point of fingers? ★ 	How do we get the customer to become more proactive 
Improve Training? Combat turnover? ★ 	How can we reduce false alarms? Maintenance Faults? ★ 	How do we predict Failure. Predictive Maintenance ★ 
Chem Cassette Improvement 	Show ECC and NDIR 	Digital Maintenance Log 
Parts replacement recommendations 	Software releases comms improvement 	PLC Wizard 
Troubleshoot multiple Vertex from one screen 	Duty Cycle Pump 	Tape and Accuracy 

Cybersecu...	Hardware and Config...	Maintenance and Reliabil...	System Integration &...	Data Managemen...	User Experi...	Operational Effi...	Training and S...
							

Steven	Paul	Mike	Chris	Tim	Ron	Radian	Sergio	Tao	Eric	Eric	Jim...	Jeremy
												

Clustering and AI synthesis

I organized all notes from the Day 1 and Day 2 workshops, first clustering them by the speaker's name. Then, I used AI to group the notes by keyword and sentiment. Each note is now tagged by both name and keyword, giving us richer context and faster access to insights. From this process, clear themes have begun to emerge, including cybersecurity, hardware and configuration, **maintenance and reliability**, system integration, data management, **user experience**, **operational efficiency**, and **training and support**.

These findings reinforce our initial pre-work and include new insights and anecdotes from FSEs explaining root causes. For example, flow rate fault alarms may result from CC wobble, while frequent downtime and support cases often stem from limited understanding of preventative maintenance.

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Additional ideas include improved **m-app documentation**, **diagnostics with recommended actions**, **clearer software versioning**, and **step-by-step guides** for optics cleaning.

Automated reporting could deliver tailored insights for different audiences, while remote troubleshooting and clearer Chemcassette visuals could reduce inefficiencies and miscommunication.

Putting it all together

I gathered all information from workshops, tech support, meetings, and post-workshop debriefs, then sorted it into 20 themes. These would be pared down further to help us prioritize the best solutions and mitigate risk.

Themes identified during the analysis centered on configuration/installation, maintenance, and user experience. Configuration themes included **software configuration**, **PLC configuration**, upgrades, **Chemcassette installation**. Maintenance themes focused on **documentation quality**, **preventative maintenance**, spare parts management, valve performance, flow rate faults, **optics issues** and **training**. Finally, user-facing themes highlighted **HMI improvements**, **specific troubleshooting recommendations**, customer downtime and opportunities for Midas/Vertex integration.

With the workshop complete and key themes identified from pre-work with technical support tickets, stakeholder interviews, and the How Might We session, we now have clear priorities and initial solution paths to pursue. Themes such as documentation, configuration, custom recommendations, preventative maintenance, and PLC issues have emerged as core areas of focus. Next, we'll examine the note clusters around these themes to determine where to concentrate our efforts.



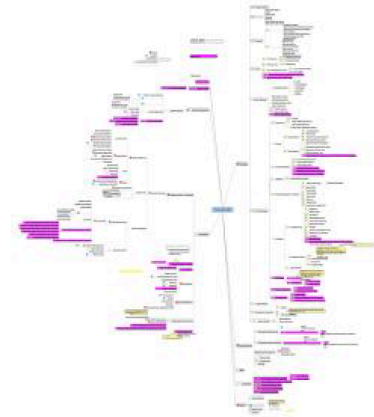
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and DOD revealed several usability and accessibility issues in their touchscreen interfaces. Poor color contrast, dense information layouts, and small touch targets make competitor screens difficult to operate and visually overwhelming.

Interpreting artifacts

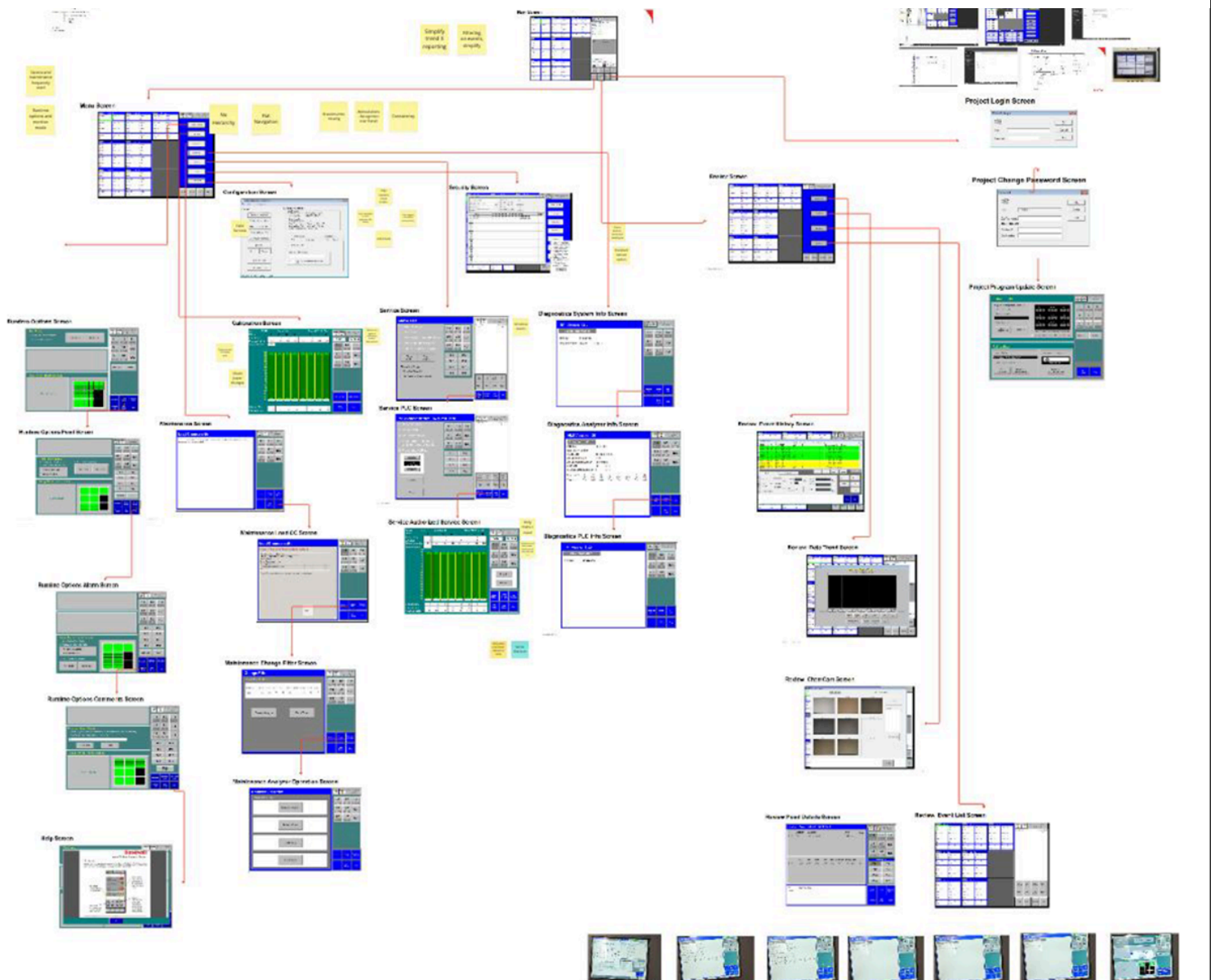
Many artifacts from the engineering team were highly technical and difficult to follow, though ultimately very informative. I analyzed and translated these materials—ranging from specifications and process diagrams to test results—into clear, shareable frameworks and visual summaries. This helped cross-functional partners quickly grasp complex concepts and apply them effectively during design and development discussions.



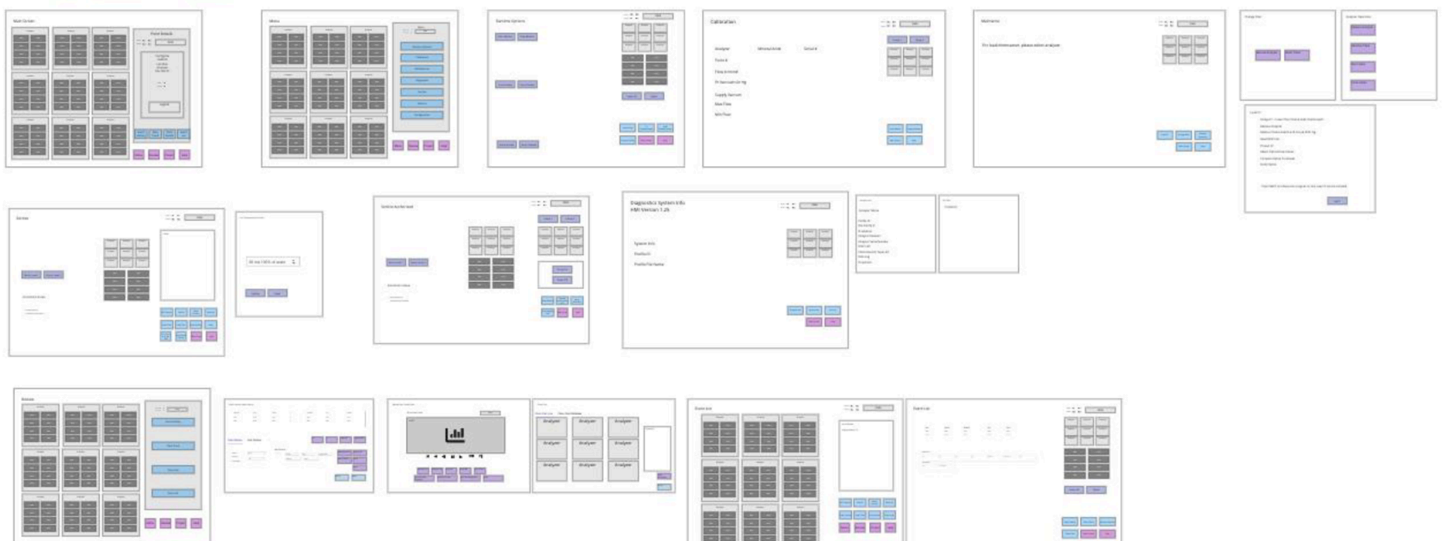
Screen Analysis

Mapping the legacy Vertex screens helped to make sense of the decades old interface. Ahead of its time, the interface was designed prior to inventions like iPhone and iPad and the advent of touch interfaces. Therefore, trial and error and a reliance on training manuals were prerequisites for usage. Wireframing the legacy interface aided my understanding of the product by removing unnecessary colors. Patterns and a system of thought began to emerge.

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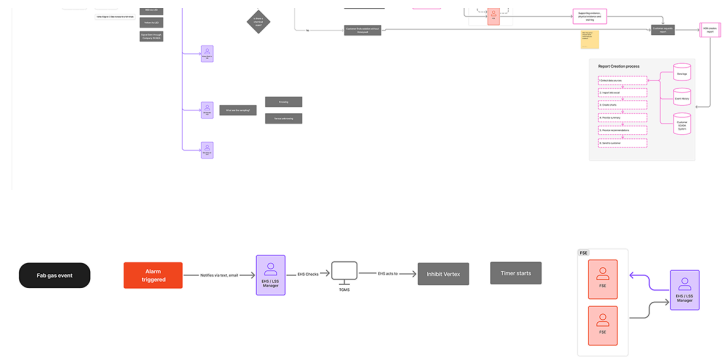
Legend



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Mapping out flows with field service helped the team to better understand the steps involved when serious issues surrounding gas detection disrupted the FAB.

Another key workflow introduced a differentiating feature for disabling auto pump swap and helped stakeholders visualize the functionality.



Wireframing

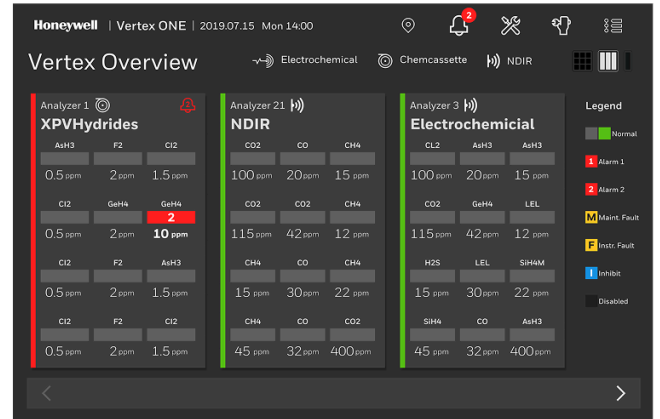
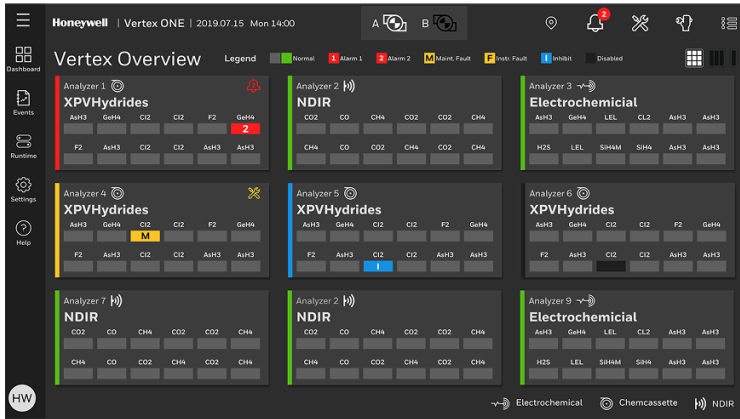
In this case wire framing really helped to reduce the noise on highly complex concepts. The engineers and design team were better able to focus on the process and the placement of key elements as we designed a pump maintenance system.

Usability testing with customers and internal stakeholders

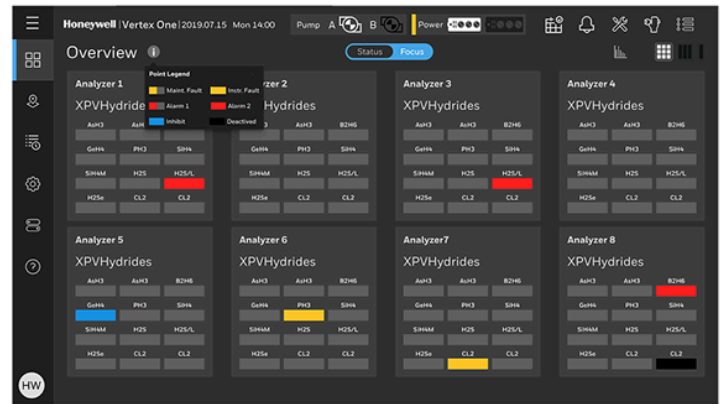
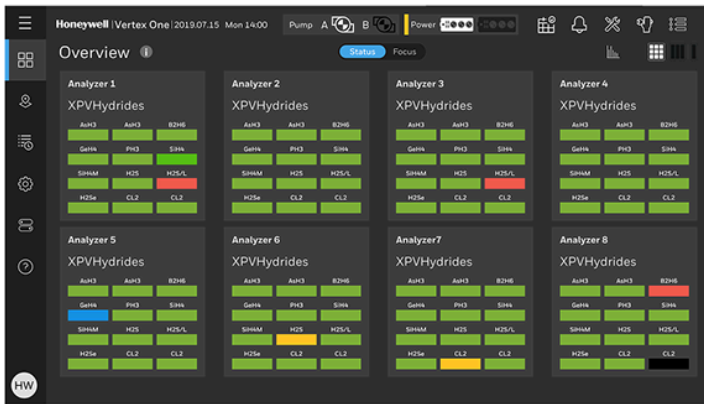
As initial concepts crystalized the team, consisting of ID, research and UX, wanted to test out the validity with customer and stakeholders. Through this process we discovered that seeing green is calming and that focus mode color palette can mean something different within the staff at the fab. We tested step-by-step guidance, and the value of an AI assistant, all of which yielded surprising results. Lastly we wanted to test the value of analytics and how the visualization would be more valuable than the current table based event history.

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attention



Interaction and screen size preference preference? HMI, tablet, remote web interface



Overview Dashboard – Status Mode

✓ Full overview with green status indicators was preferred.



Optional Icons – Assist Mode

✓ Can be useful for training new hires, but would have to be disabled later

Overview Dashboard – Focus Mode

⚠ Gray points perceived as disabled or turned off. "Sleep mode" idea.

✓ Toggle option can help to switch modes based on user preference.

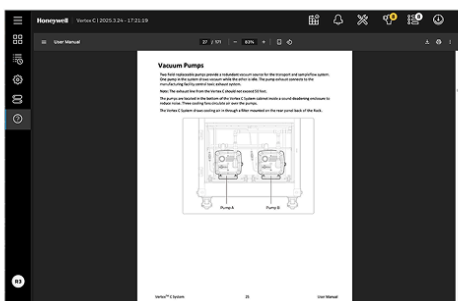
Status and Focus Toggle



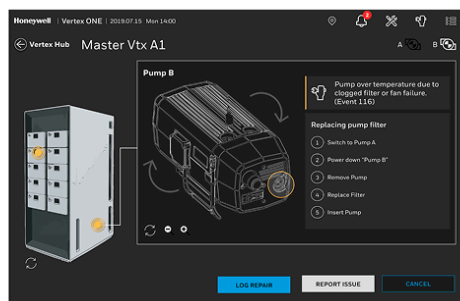
Need consistency in color representation across different interfaces

Color Indicators	Meaning
Green	Active
Red	Inactive or Faulty
Grayed Out	Disabled/Inhibited

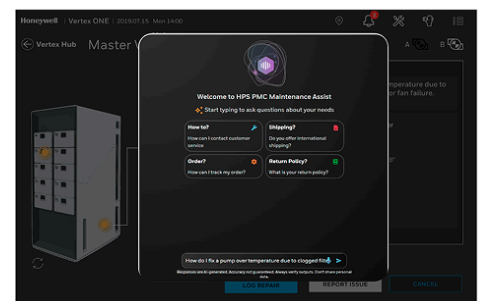
0 User Manual



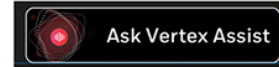
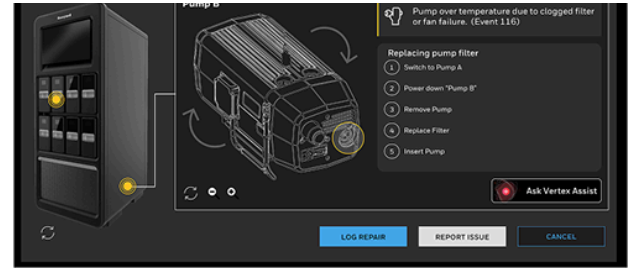
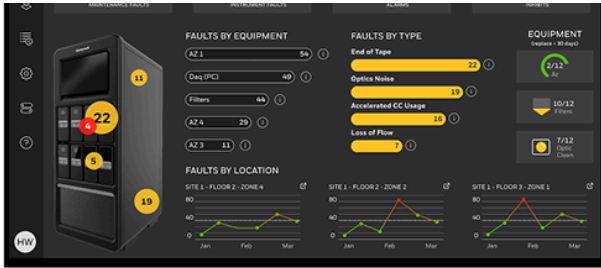
1 Visual step-by-guidance



2 Maintenance Assist on the Edge



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Data analysis performed by HON Field Services for quarterly business reviews with SAS:

- Pareto chart for faults of all the HON equipment on site
- Types of faults
- Fault response time
- Uptime & downtime breakdown

Maintenance Assist on the Edge

⚠ Connectivity disabled for any Cloud-based solution

✅ Integration with Salesforce will add value to track LSS/FSE tickets

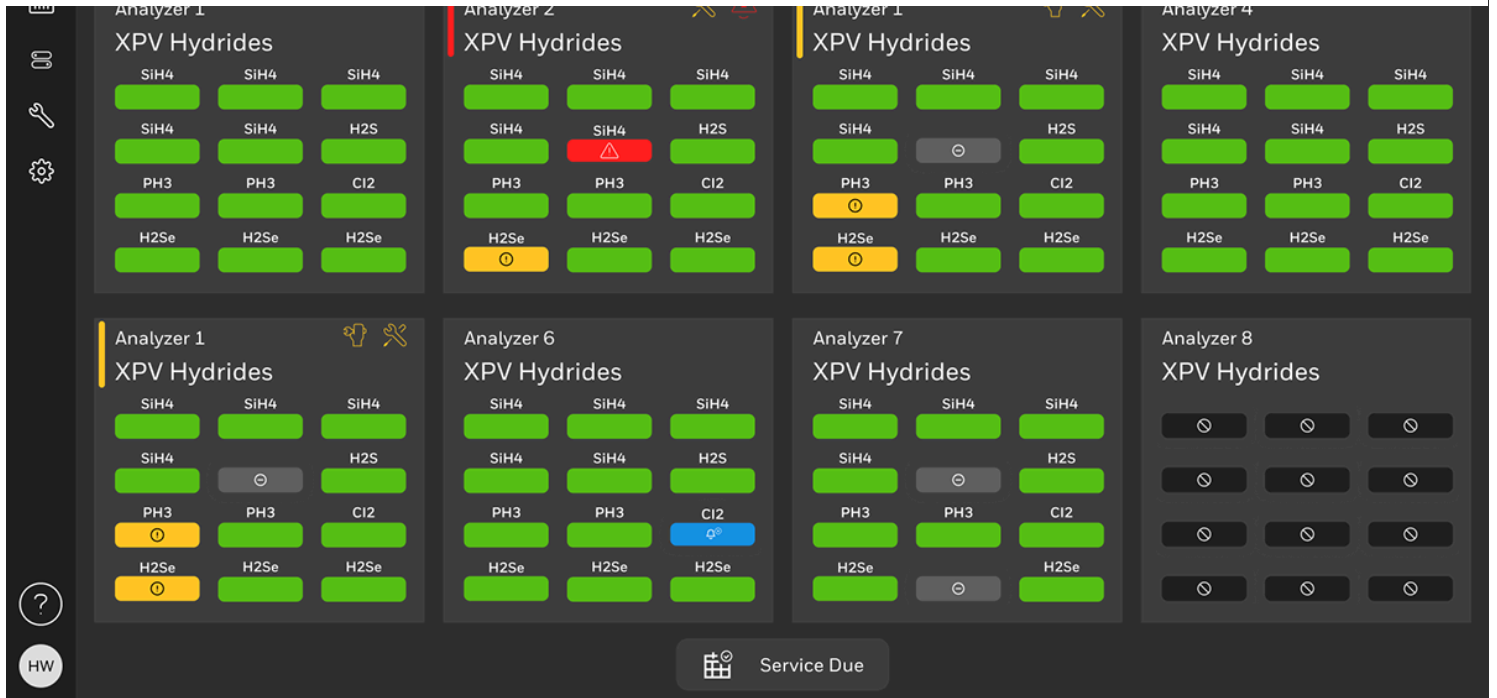
Finalizing Design Screens

After thoroughly and meticulously researching the new Vertex UI, synthesizing and theming the result. After the interviews, VOC, usability studies, prototypes and wireframing ... only then am I ready for finalizing the user interface. This feels like the easy part after all of the preparation. My method is for user interface design relies on research, screen comparisons, and brand considerations. I want to use the latest design system that the Vertex seems modern and there is also special consideration for the touch screen nature of the HMI and the fact that people may be looking up at the interface. I considered alarms, complicated maintenance scenarios and the latest trends in AI and digital twins.

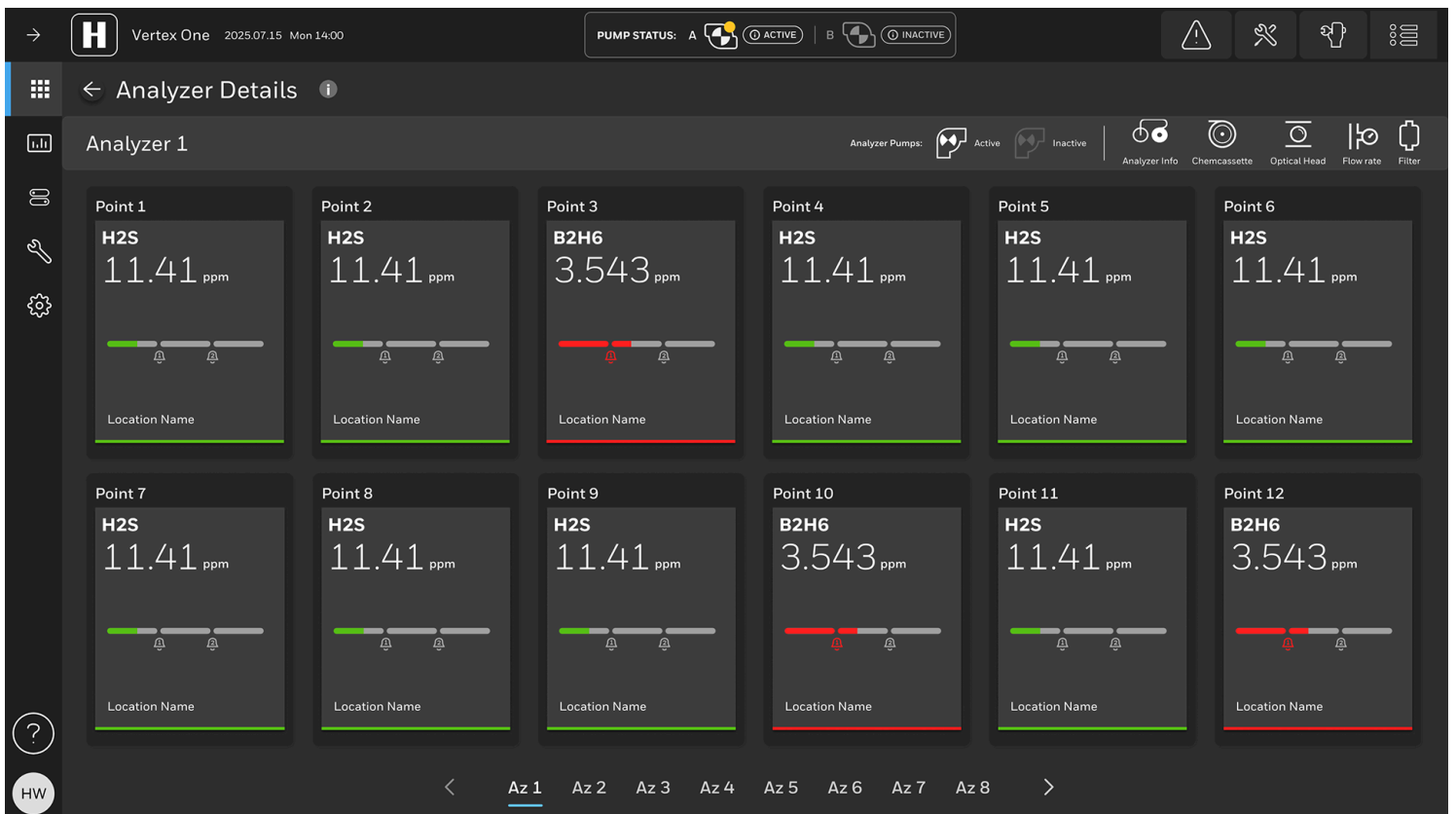
Overview screens and event panels

The new scan line mirrors the exterior indication of alarms and alerts. The icons within alarm and faults on each point is a special consideration for low vision and color blind users as red and green is a common challenge for color blind users. We made the pump status more easy to digest and more descriptive of faults since this is one of the differentiators the new Vertex. Location grouping was a submitted invention disclosure for this project as was another feature for disabling the automatic pump swap.

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Listening to feedback

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stadium when purchasing a ticket and mapping that to the number of people attending. What options do they want with for their seats and who should be notified.

→

H

Vertex One 2025.07.15 Mon 14:00

☰

← New Profile ⓘ

Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 04	Card 05	Card 06	Card 07
Slot 01	MAINT	AL2	AL2	AL2
Slot 02	BOTH	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL2	AL1	AL1	AL1
	Card 08	Card 09	Card 10	Card 11
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
	Card 12	Card 13	Card 14	Card 15
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items

<

4-15

16-27

28-39

40-45

>

Nothing selected yet

Select a card relay first

Cancel

Save As

Install

Save

?

HW

[illegible]

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Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 01	Card 02	Card 03	Card 04
Slot 01	MAINT	MAINT	MAINT	MAINT
Slot 02	INSTR	INSTR	INSTR	BOTH
Slot 04	MAINT	MAINT	MAINT	AL1
Slot 04	INSTR	INSTR	INSTR	AL2

Select analyzer points to map

☒ Analyzer 01

☐ Analyzer 02

☒ Analyzer 03

☐ Analyzer 04

☐ Analyzer 05

☐ Analyzer 06

☒ Analyzer 07

☐ Analyzer 08

Select Fault Type *

☒ Maintenance Fault ☐ Instrument Fault ☐ Both ☐ None

Cancel

Apply

Cancel

Save As

Install

Save

→

H

 Vertex One 2025.07.15 Mon 14:00

← New Profile

Analyzer Details

Service Due

Events

Timeouts

Serial Comm

Database

PLC Config

Location

Presets

Fault Relay Mappings

Alarm Relay Mappings

	Card 04	Card 05	Card 06	Card 07
Slot 01	MAINT	AL2	AL2	AL2
Slot 02	BOTH	AL2	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL2	AL1	AL1	AL1

	Card 08	Card 09	Card 10	Card 11
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

	Card 12	Card 13	Card 14	Card 15
Slot 01	AL2	AL2	AL2	AL2
Slot 02	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1
Slot 04	AL1	AL1	AL1	AL1

4-15 of 45 items

< 4-15 16-27 28-39 40-45 >

Select analyzer points to map

Analyzer 01

Analyzer 02

Analyzer 03

Analyzer 04

Analyzer 05

Analyzer 06

Analyzer 07

Analyzer 08

Select Alarm Type *

☐ Alarm 1 ☒ Alarm 2 ☐ Both ☐ None

Cancel

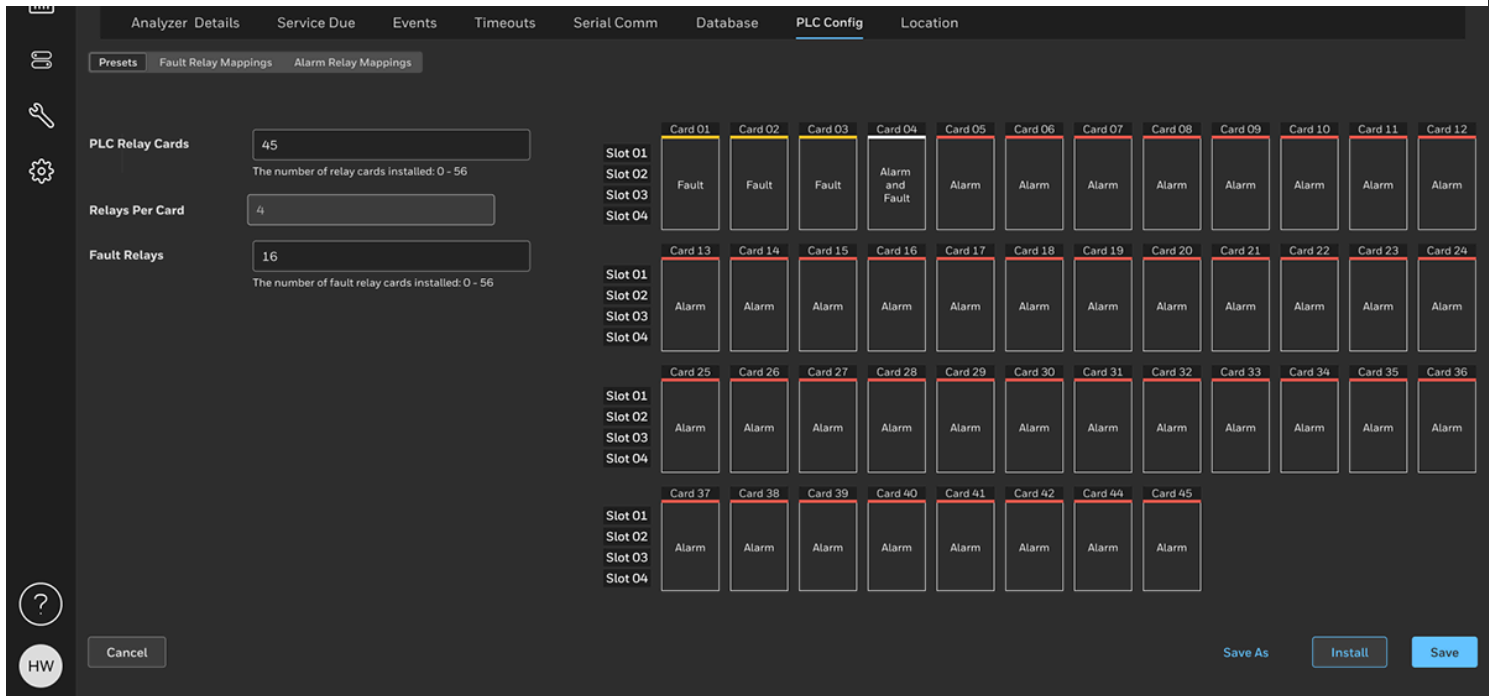
Apply

Save

Save As

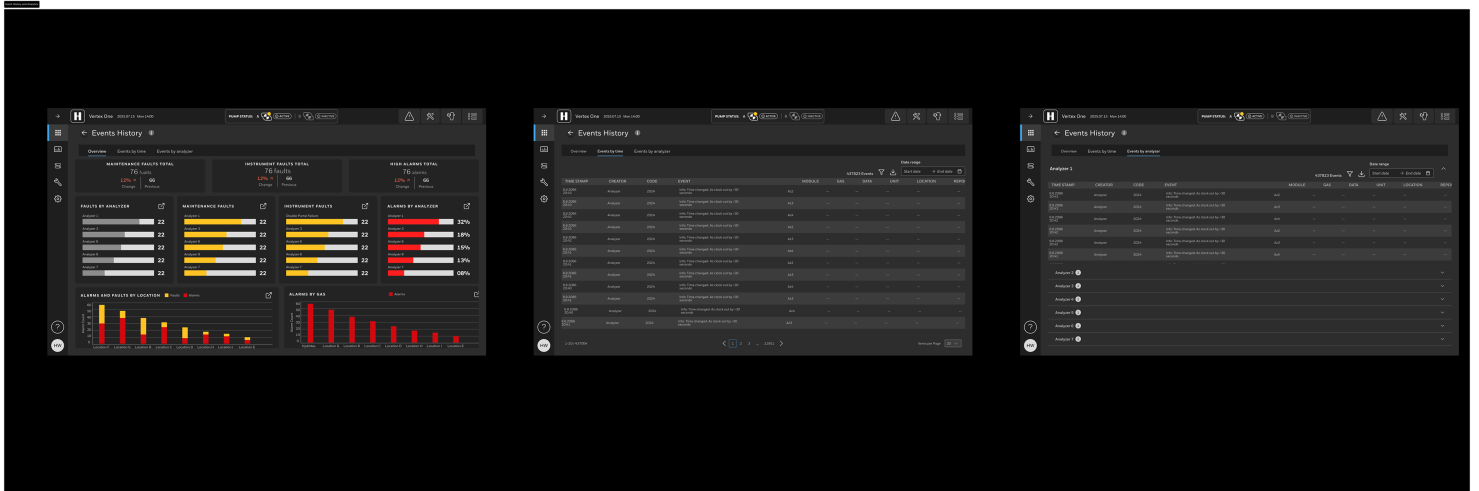
Install

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Analytics and Event History

We also heard that field service technicians wanted more insights into faults and alarms on the Vertex. So we provided easy to digest visualizations that also helped with the quarterly reporting process for customers. The resulting savings could amount to \$1M+ annually and increase revenue for Honeywell in the high tech space.



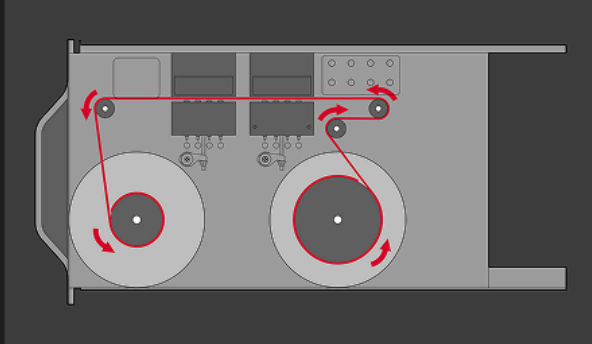
Training and guided maintenance

We heard repetitively that documentation was lacking and that training was needed. We solved this by explicitly detailing the steps of complex tasks like replacing Chemcassettes while providing instant and

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Please open each analyzer and replace the chemcassette.
When the all chemcassettes are placed, please click NEXT.



Cancel

Next

- ✓ Release analyzer
- ✓ Replace Chemcassette
- Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle

Replace Chemcassette: Analyzer 1



Uploading data from the chemcassettes RFID tag



- ✓ Release analyzer
- ✓ Replace Chemcassette
- ✓ Uploading data from RFID
- Verify Chemcassette status
- Adjust white, light/dark gray
- Resume monitor or idle

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i

Information of the new chemcassettes

Analyzer 1
Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 7
Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 3
Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 8
Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

Analyzer 5
Gas family: XPVHydrides
Serial no.: 123456789
Tape status: Unused
Expiry date: Jun 3, 2026

✓ Release analyzer

✓ Replace Chemcassette

✓ Uploading data from RFID

○ Verify Chemcassette status

• Adjust white, light/dark gray

• Resume monitor or idle

Cancel

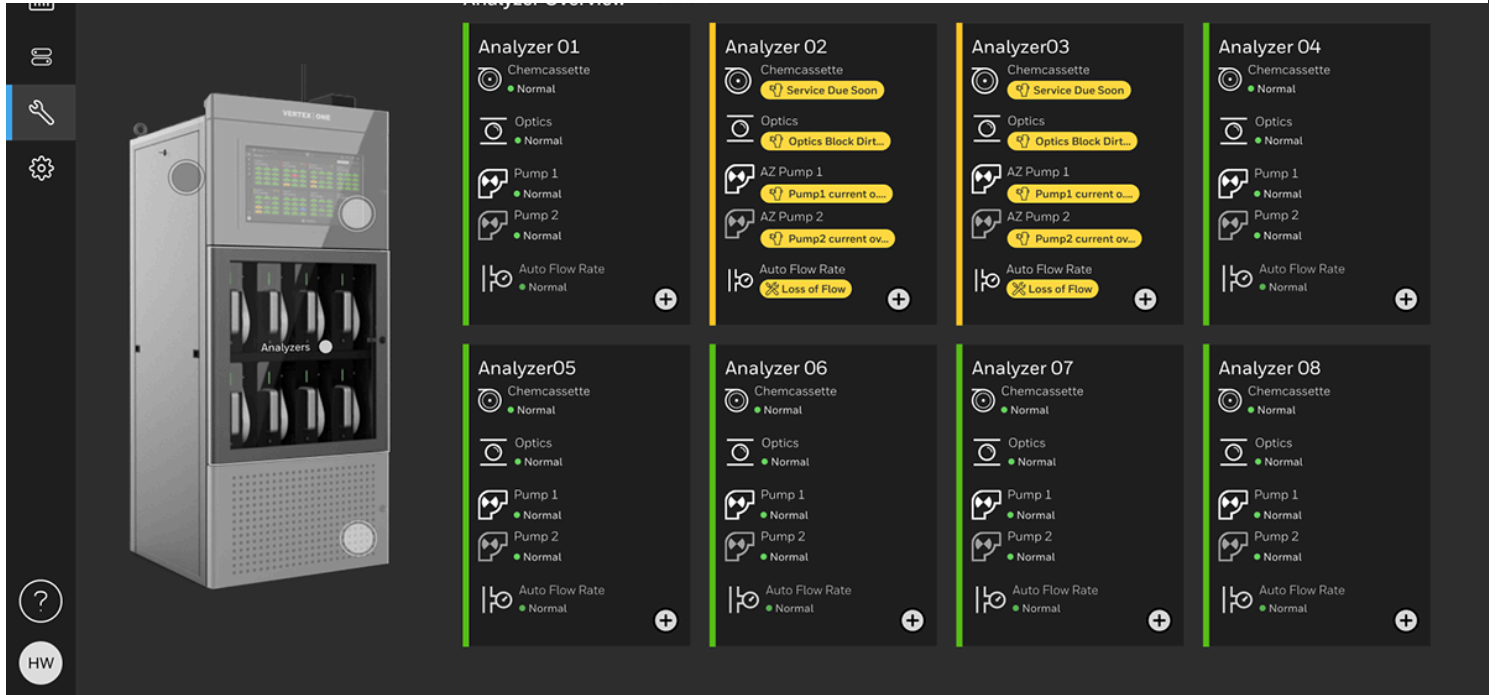
Next

Diagnostics and Recommendations

Ideas circulated around specific recommendations with the rise of artificial intelligence. Internal stakeholders loved this feature including FSEs with many years of experience. The recommendations synthesized from various data sources could be a game changer for Honeywell and allow technicians to spend more time with the customer and less time figuring out how to troubleshoot edge case issues. The digital twin aspect of the diagnostics section received applause from sales, marketing and technicians.

"With the analytics and diagnostics ... This is something we can sell" - Honeywell High Tech President

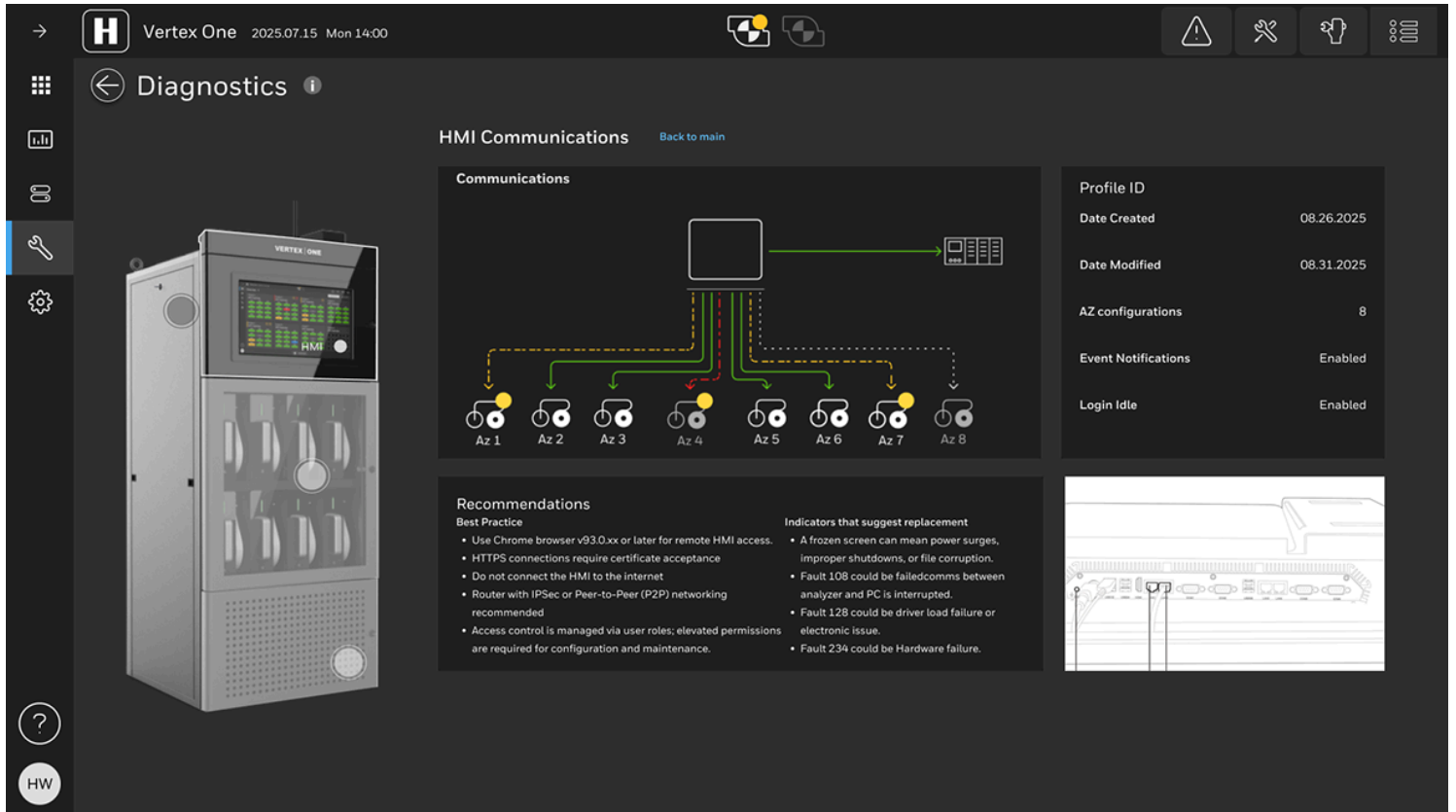
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The main screen of the Vertex One HMI displays a 2x4 grid of 8 analyzer status cards. Each card shows the following components and their status:

- Analyzer 01:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).
- Analyzer 02:** Chemcassette (Normal), Optics (Normal), AZ Pump 1 (Pump1 current o...), AZ Pump 2 (Pump2 current ov...), Auto Flow Rate (Loss of Flow).
- Analyzer 03:** Chemcassette (Normal), Optics (Normal), AZ Pump 1 (Pump1 current o...), AZ Pump 2 (Pump2 current ov...), Auto Flow Rate (Loss of Flow).
- Analyzer 04:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).
- Analyzer 05:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).
- Analyzer 06:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).
- Analyzer 07:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).
- Analyzer 08:** Chemcassette (Normal), Optics (Normal), Pump 1 (Normal), Pump 2 (Normal), Auto Flow Rate (Normal).

Each card has a green status indicator and a '+' button in the bottom right corner.



The Diagnostics screen provides detailed information about the HMI's communication and configuration. It includes a diagram of the HMI's internal components, a list of recommendations, and a profile ID section.

HMI Communications

Back to main

Communications

The diagram shows the HMI's internal components and their connections. The HMI is connected to 8 analyzers (Az 1 to Az 8) via a central communication bus. The connections are color-coded: green for normal, yellow for warning, and red for error.

Recommendations

Best Practice

- Use Chrome browser v93.0.xx or later for remote HMI access.
- HTTPS connections require certificate acceptance
- Do not connect the HMI to the internet
- Router with IPSec or Peer-to-Peer (P2P) networking recommended
- Access control is managed via user roles; elevated permissions are required for configuration and maintenance.

Indicators that suggest replacement

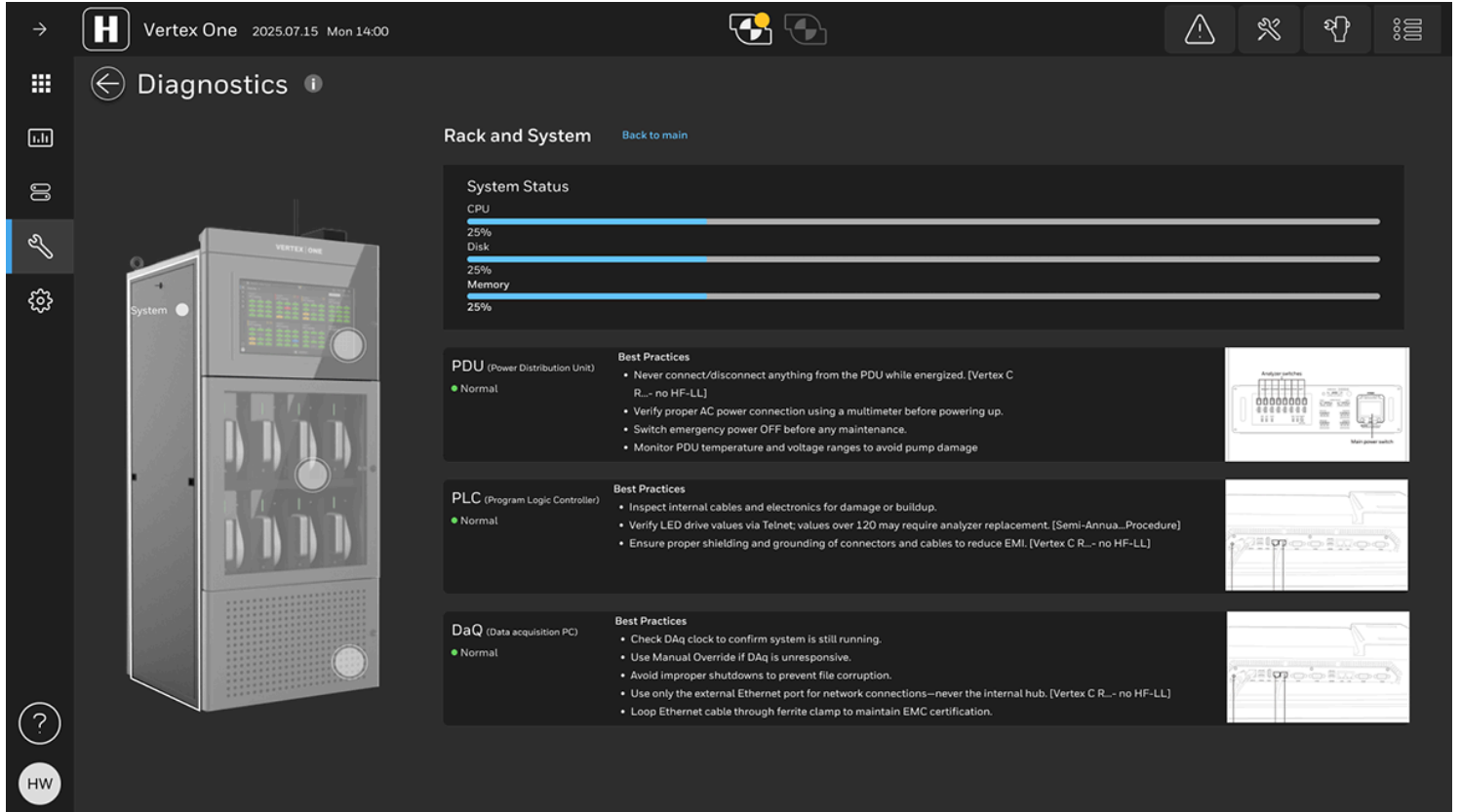
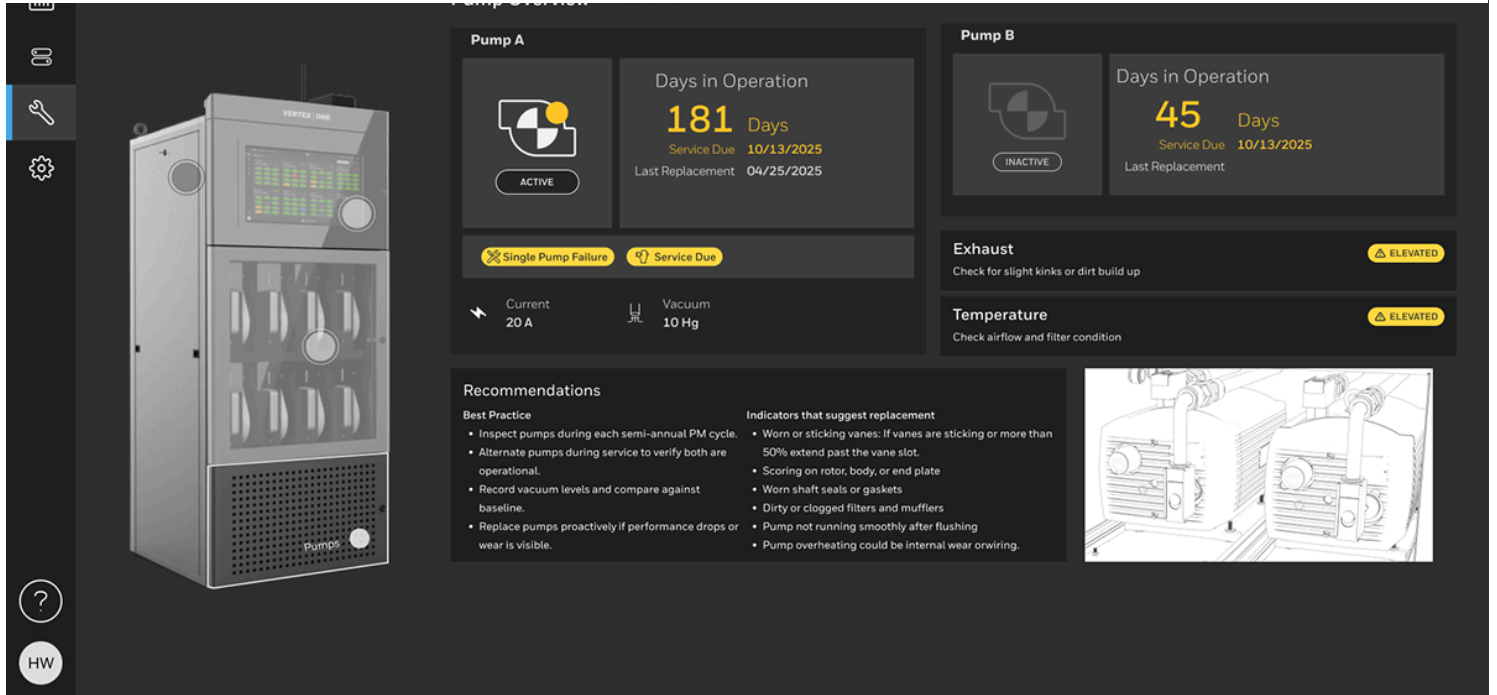
- A frozen screen can mean power surges, improper shutdowns, or file corruption.
- Fault 108 could be failedcomms between analyzer and PC is interrupted.
- Fault 128 could be driver load failure or electronic issue.
- Fault 234 could be Hardware failure.

Profile ID

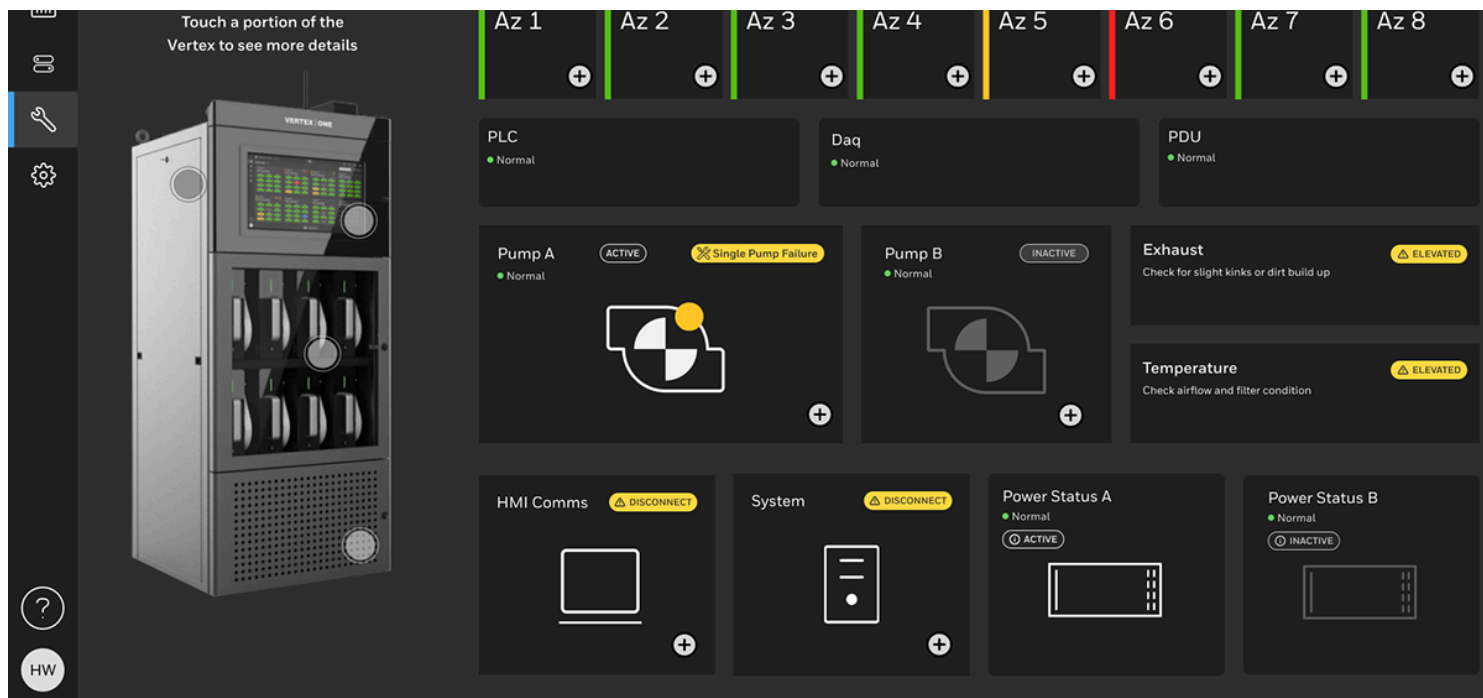
Field	Value
Date Created	08.26.2025
Date Modified	08.31.2025
AZ configurations	8
Event Notifications	Enabled
Login Idle	Enabled

The bottom right corner of the screen shows a technical drawing of the HMI's internal components, including the display, keyboard, and various ports.

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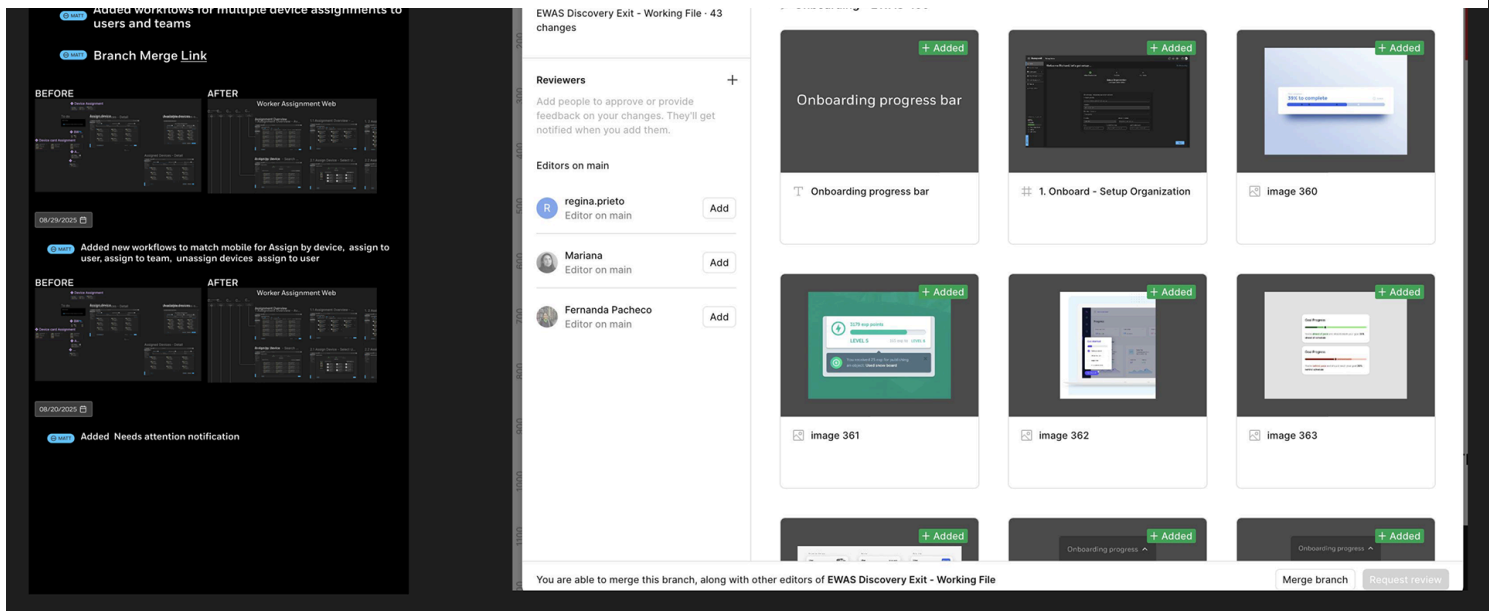
Prototype

The prototype helped to test the final screens with users and internal stakeholders alike. Every screen was linked up to the prototype and we were able to hook it into a tablet that mimicked the actual Vertex machine.

Versioning, Branching and Annotating

During the dev handoff process it's critical to keep designs updated when there will inevitably be minor tweaks. This is why I implemented a versioning system that included branch and merge functionality similar to GitHub. This native-Figma functionality allowed designers and developers to stay in sync and schedule updates after release in subsequent sprints. Annotating designs is another critical and often overlooked part of the design process. Annotations reduce the amount of back and forth between developers and stakeholders alike.

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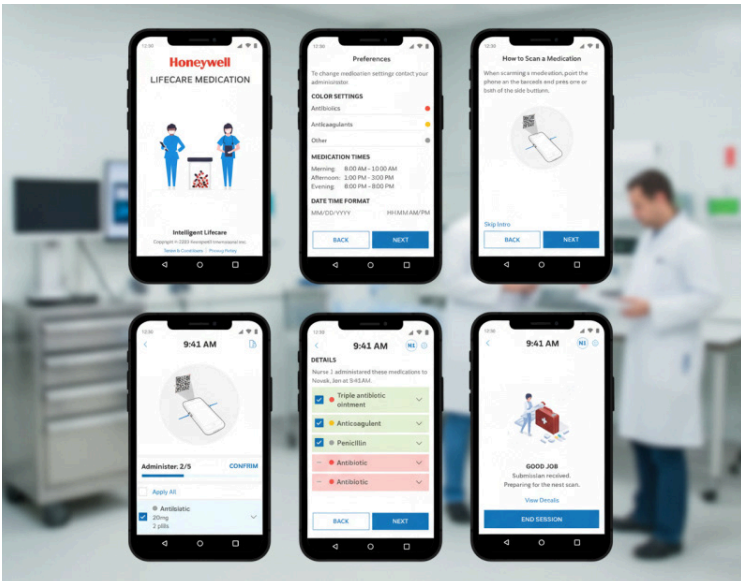


Keep going and going ...

Successful products and upgrades don't happen overnight or even on the first release. There will always be feedback that designers didn't consider or hear about until after the release. There will be more ways to increase revenue after each successful release as customers see tremendous value in premium features.

That's the idea behind the new Vertex. Not only do we have plans to release an integration with customer application ecosystems but we also plan to introduce remote management and more AI-based features like more predictive analytics and integrating with Honeywell's gas portfolio and seeing where synergies can lie between Honeywell's extremely diverse product offerings.

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